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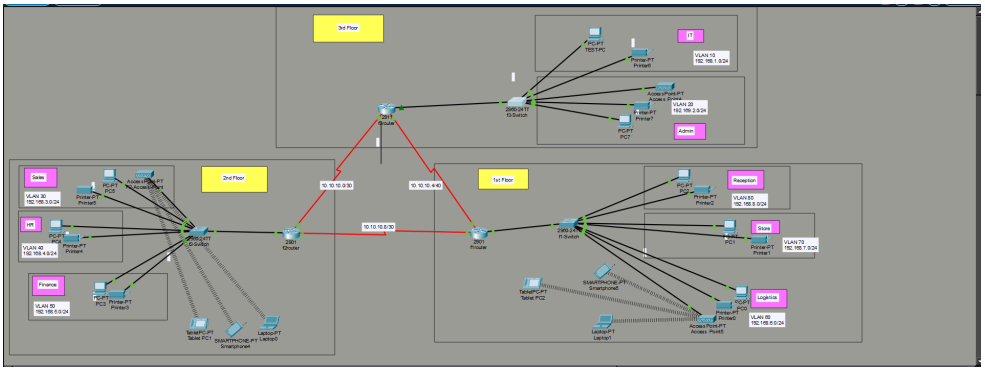
Experiment No.	10

AIM:	To design and implement a multi-floor hotel network using VLANs, Inter-VLAN Routing, DHCP, OSPF routing, SSH, and Port Security in Cisco Packet Tracer.
Program 1	
Theory:	Cisco Packet Tracer, developed by Cisco Systems, is a network simulation tool used to design, configure, and analyze network topologies without requiring physical hardware. In this experiment, a multi-floor hotel network was designed consisting of three floors, each having multiple departments. Each department was assigned a separate VLAN to logically segment the network. The key networking concepts implemented in this experiment include: ● VLAN (Virtual LAN): Used to divide the network into logical segments for departments. ● Inter-VLAN Routing: Configured using router sub-interfaces to allow communication between VLANs. ● DHCP (Dynamic Host Configuration Protocol): Used to automatically assign IP addresses to devices. ● OSPF (Open Shortest Path First): A dynamic routing protocol used to exchange routing information between routers. ● SSH (Secure Shell): Used for secure remote login to routers. ● Port Security: Used to restrict access to specific switch ports. Each floor consists of: ● One router ● One switch ● Multiple departments (VLANs) ● Wireless access point ● PCs and printers

	Routers are interconnected using serial DCE cables, and clock rate is configured on DCE interfaces. The network ensures full communication between all devices across different VLANs and floors.
PROBLEM STATEMENT :	**.Task 1: Design Network Topology** **Problem:** Design a hotel network with 3 floors and multiple departments. **How we did:** ● Created 3 routers (one per floor) ● Connected routers using serial DCE cables ● Added switches, PCs, printers, and access points ● Divided floors into departments  **Task 2: VLAN Configuration** **Problem:** Assign each department to a separate VLAN. **How we did:** **First Floor:** ● VLAN 80 → Reception → 192.168.8.0 ● VLAN 70 → Store → 192.168.7.0 ● VLAN 60 → Logistics → 192.168.6.0

Second Floor:

- VLAN 50 → Finance → 192.168.5.0
- VLAN 40 → HR → 192.168.4.0
- VLAN 30 → Sales → 192.168.3.0

Third Floor:

- VLAN 20 → Admin → 192.168.2.0
- VLAN 10 → IT → 192.168.1.0

Configured VLANs on switches and assigned ports accordingly.
Configured trunk ports for router communication.

F1-Switch:

```
Switch#show vlan brief

VLAN Name                             Status    Ports
---- -------------------------------- --------- -------------------------------
1    default                          active    Fa0/8, Fa0/9, Fa0/10, Fa0/11
                                                Fa0/12, Fa0/13, Fa0/14, Fa0/15
                                                Fa0/16, Fa0/17, Fa0/18, Fa0/19
                                                Fa0/20, Fa0/21, Fa0/22, Fa0/23
                                                Fa0/24, Gig0/1, Gig0/2
60   VLAN0060                         active    Fa0/5, Fa0/7
70   VLAN0070                         active    Fa0/2, Fa0/6
80   VLAN0080                         active    Fa0/3, Fa0/4
```

```
interface FastEthernet0/2
 switchport access vlan 70
 switchport mode access
!
interface FastEthernet0/3
 switchport access vlan 80
 switchport mode access
!
interface FastEthernet0/4
 switchport access vlan 80
 switchport mode access
!
interface FastEthernet0/5
 switchport access vlan 60
 switchport mode access
!
 --More--
```

F2-Switch:

```
Switch#
Switch#show vlan brief

VLAN Name                             Status    Ports
---- -------------------------------- --------- -------------------------------
1    default                          active    Fa0/8, Fa0/9, Fa0/10, Fa0/11
                                                Fa0/12, Fa0/13, Fa0/14, Fa0/15
                                                Fa0/16, Fa0/17, Fa0/18, Fa0/19
                                                Fa0/20, Fa0/21, Fa0/22, Fa0/23
                                                Fa0/24, Gig0/1, Gig0/2
30   VLAN0030                         active    Fa0/3, Fa0/6
40   VLAN0040                         active    Fa0/2
50   VLAN0050                         active    Fa0/4, Fa0/5, Fa0/7
1002 fddi-default                     active
```

```
!
spanning-tree mode pvst
spanning-tree extend system-id
!
interface FastEthernet0/1
 switchport mode trunk
!
interface FastEthernet0/2
 switchport access vlan 40
 switchport mode access
!
interface FastEthernet0/3
 switchport access vlan 30
 switchport mode access
!
interface FastEthernet0/4
 switchport access vlan 50
 switchport mode access
!
 --More--
```

F3-Switch:

```
Switch#show vlan brief

VLAN Name                             Status    Ports
---- -------------------------------- --------- -------------------------------
1    default                          active    Fa0/6, Fa0/7, Fa0/8, Fa0/9
                                                Fa0/10, Fa0/11, Fa0/12, Fa0/13
                                                Fa0/14, Fa0/15, Fa0/16, Fa0/17
                                                Fa0/18, Fa0/19, Fa0/20, Fa0/21
                                                Fa0/22, Fa0/23, Fa0/24, Gig0/1
                                                Gig0/2
10   VLAN0010                         active    Fa0/2, Fa0/5
20   VLAN0020                         active    Fa0/3, Fa0/4
```

```
!
!
spanning-tree mode pvst
spanning-tree extend system-id
!
interface FastEthernet0/1
 switchport mode trunk
!
interface FastEthernet0/2
 switchport access vlan 10
 switchport mode access
 switchport port-security
 switchport port-security mac-address sticky
!
interface FastEthernet0/3
 switchport access vlan 20
 switchport mode access
```

Task 3: Configure Router Interfaces

Problem:
Assign IP addresses to router interfaces.

How we did:

- Configured serial interfaces with /30 networks
- Enabled interfaces using no shutdown
- Configured clock rate (64000) on DCE interfaces

Example:

interface serial 0/2/0

ip address 10.10.10.5 255.255.255.252

clock rate 64000

no shutdown

F1-Switch:

```
Switch#show interfaces trunk
Port        Mode         Encapsulation  Status        Native vlan
Fa0/1       on           802.1q         trunking      1

Port        Vlans allowed on trunk
Fa0/1       1-1005

Port        Vlans allowed and active in management domain
Fa0/1       1,60,70,80

Port        Vlans in spanning tree forwarding state and not pruned
Fa0/1       1,60,70,80

Switch#
```

F2-Switch:

```
Switch#
Switch#show interfaces trunk
Port        Mode         Encapsulation  Status        Native vlan
Fa0/1       on           802.1q         trunking      1

Port        Vlans allowed on trunk
Fa0/1       1-1005

Port        Vlans allowed and active in management domain
Fa0/1       1,30,40,50

Port        Vlans in spanning tree forwarding state and not pruned
Fa0/1       1,30,40,50

Switch#
```

F3-Switch:

```
Switch#
Switch#show interfaces trunk
Port        Mode         Encapsulation  Status        Native vlan
Fa0/1       on           802.1q         trunking      1

Port        Vlans allowed on trunk
Fa0/1       1-1005

Port        Vlans allowed and active in management domain
Fa0/1       1,10,20

Port        Vlans in spanning tree forwarding state and not pruned
Fa0/1       1,10,20

Switch#
```

Task 4: Inter-VLAN Routing

Problem:
Enable communication between VLANs.

How we did:

- Created sub-interfaces on routers
- Assigned VLAN IDs using encapsulation

Example:

interface g0/0.80

encapsulation dot1Q 80

ip address 192.168.8.1 255.255.255.0

Each sub-interface acts as a **default gateway**.

F1-Router:

```
F1-router>en
F1-router#show ip interface brief
Interface              IP-Address      OK? Method Status                Protocol
GigabitEthernet0/0     unassigned      YES NVRAM  up                    up
GigabitEthernet0/0.60  192.168.6.1     YES manual up                    up
GigabitEthernet0/0.70  192.168.7.1     YES manual up                    up
GigabitEthernet0/0.80  192.168.8.1     YES manual up                    up
GigabitEthernet0/1     unassigned      YES NVRAM  administratively down down
Serial0/3/0            10.10.10.2      YES manual up                    up
Serial0/3/1            10.10.10.6      YES manual up                    up
Vlan1                  unassigned      YES unset  administratively down down
F1-router#
```

F2-Router:

```
F2-router>en
F2-router#show ip interface brief
Interface              IP-Address      OK? Method Status                Protocol
GigabitEthernet0/0     unassigned      YES NVRAM  up                    up
GigabitEthernet0/0.30  192.168.3.1     YES manual up                    up
GigabitEthernet0/0.40  192.168.4.1     YES manual up                    up
GigabitEthernet0/0.50  192.168.5.1     YES manual up                    up
GigabitEthernet0/1     unassigned      YES NVRAM  administratively down down
Serial0/3/0            10.10.10.6      YES manual up                    up
Serial0/3/1            10.10.10.10     YES manual up                    up
Vlan1                  unassigned      YES unset  administratively down down
F2-router#
```

F3-Router:

```
F3-Router>en
F3-Router#show ip interface brief
Interface              IP-Address      OK? Method Status                Protocol
GigabitEthernet0/0     unassigned      YES NVRAM  up                    up
GigabitEthernet0/0.10  192.168.1.1     YES manual up                    up
GigabitEthernet0/0.20  192.168.2.1     YES manual up                    up
GigabitEthernet0/1     unassigned      YES NVRAM  administratively down down
GigabitEthernet0/2     unassigned      YES NVRAM  administratively down down
Serial0/3/0            10.10.10.5      YES manual up                    up
Serial0/3/1            10.10.10.1      YES manual up                    up
Vlan1                  unassigned      YES unset  administratively down down
F3-Router#
```

Task 5: DHCP Configuration

Problem:
Assign IP addresses dynamically.

How we did:
Configured DHCP on routers:

ip dhcp pool RECEPTION

network 192.168.8.0 255.255.255.0

default-router 192.168.8.1

dns-server 192.168.8.1

Created separate pools for each VLAN.

F1-Router:

```
F1-router#show running-config | section GigabitEthernet0/0
interface GigabitEthernet0/0
 no ip address
 duplex auto
 speed auto
interface GigabitEthernet0/0.60
 encapsulation dot1Q 60
 ip address 192.168.6.1 255.255.255.0
interface GigabitEthernet0/0.70
 encapsulation dot1Q 70
 ip address 192.168.7.1 255.255.255.0
interface GigabitEthernet0/0.80
 encapsulation dot1Q 80
 ip address 192.168.8.1 255.255.255.0
F1-router#
```

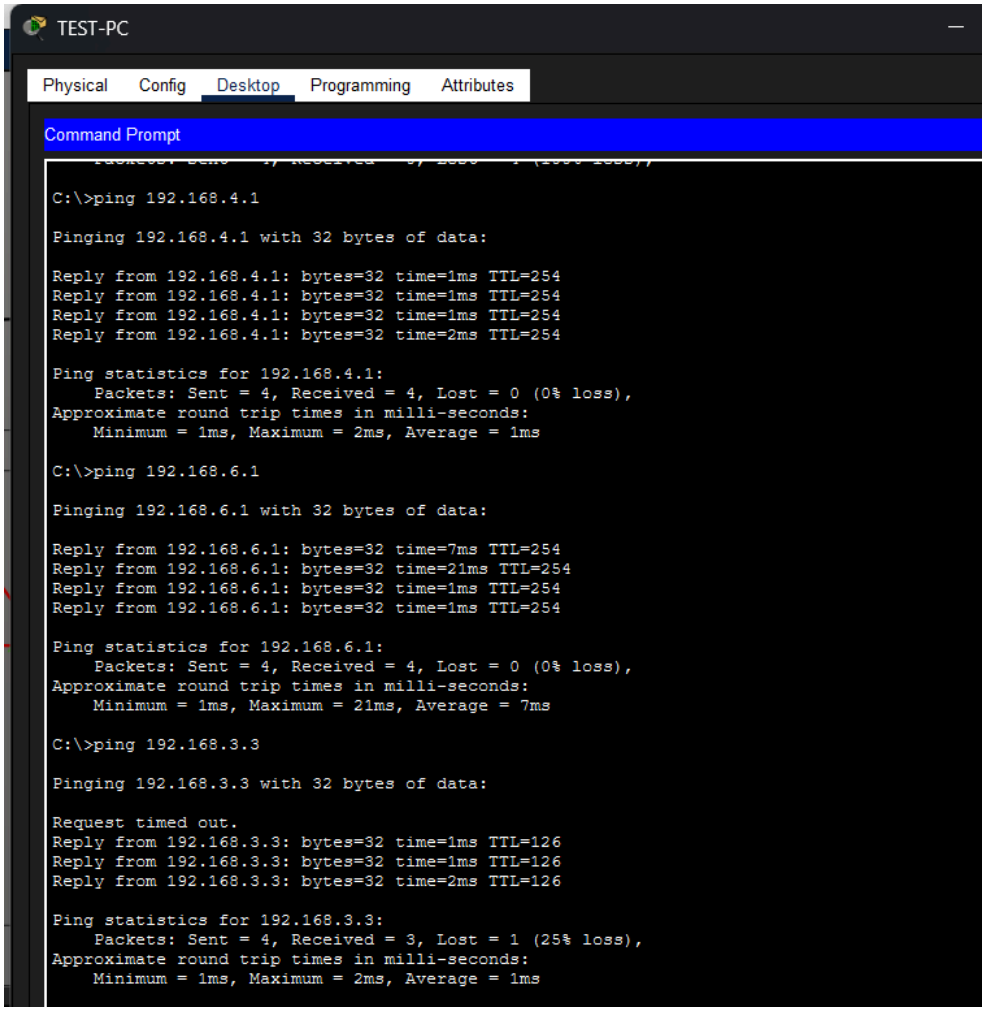
F2-Router:

```
F2-router#show running-config | section GigabitEthernet0/0
interface GigabitEthernet0/0
 no ip address
 duplex auto
 speed auto
interface GigabitEthernet0/0.30
 encapsulation dot1Q 30
 ip address 192.168.3.1 255.255.255.0
interface GigabitEthernet0/0.40
 encapsulation dot1Q 40
 ip address 192.168.4.1 255.255.255.0
interface GigabitEthernet0/0.50
 encapsulation dot1Q 50
 ip address 192.168.5.1 255.255.255.0
F2-router#
```

F3-Router:

```
F3-Router#show running-config | section GigabitEthernet0/0
interface GigabitEthernet0/0
 no ip address
 duplex auto
 speed auto
interface GigabitEthernet0/0.10
 encapsulation dot1Q 10
 ip address 192.168.1.1 255.255.255.0
interface GigabitEthernet0/0.20
 encapsulation dot1Q 20
 ip address 192.168.2.1 255.255.255.0
```

Ping between VLANs:



Task 6: Configure OSPF Routing

Problem:

Enable communication between different floors.

How we did:

router ospf 10

network 10.10.10.0 0.0.0.3 area 0

network 192.168.x.0 0.0.0.255 area 0

- Advertised all networks
- Used Area 0

DHCP configuration

F1-Router:

```
 ip address 192.168.6.1 255.255.255.0
F1-router#show running-config | section dhcp
ip dhcp excluded-address 192.168.8.1
ip dhcp excluded-address 192.168.7.1
ip dhcp excluded-address 192.168.6.1
ip dhcp pool Reception
 network 192.168.8.0 255.255.255.0
 default-router 192.168.8.1
 dns-server 192.168.8.1
ip dhcp pool Store
 network 192.168.7.0 255.255.255.0
 default-router 192.168.7.1
 dns-server 192.168.7.1
ip dhcp pool Logistics
 network 192.168.6.0 255.255.255.0
 default-router 192.168.6.1
 dns-server 192.168.6.1
F1-router#
```

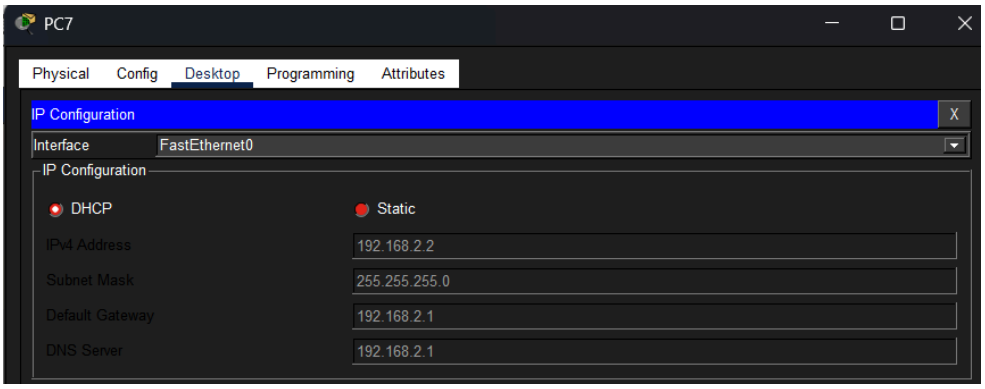
F2-Router:

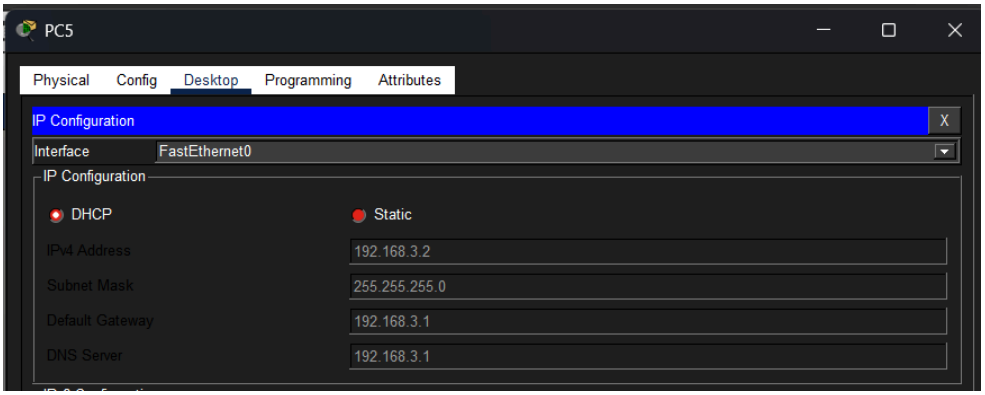
```
F2-router#show running-config | section dhcp
ip dhcp pool Finance
 network 192.168.5.0 255.255.255.0
 default-router 192.168.5.1
 dns-server 192.168.5.1
ip dhcp pool Sales
 network 192.168.3.0 255.255.255.0
 default-router 192.168.3.1
 dns-server 192.168.3.1
ip dhcp pool HR
 network 192.168.4.0 255.255.255.0
 default-router 192.168.4.1
 dns-server 192.168.4.1
F2-router#
```

F3-Router:

```
 ip address 192.168.2.1 255.255.255.0
F3-Router#show running-config | section dhcp
ip dhcp pool IT
 network 192.168.1.0 255.255.255.0
 default-router 192.168.1.1
 dns-server 192.168.1.1
ip dhcp pool Admin
 network 192.168.2.0 255.255.255.0
 default-router 192.168.2.1
 dns-server 192.168.2.1
F3-Router#
```

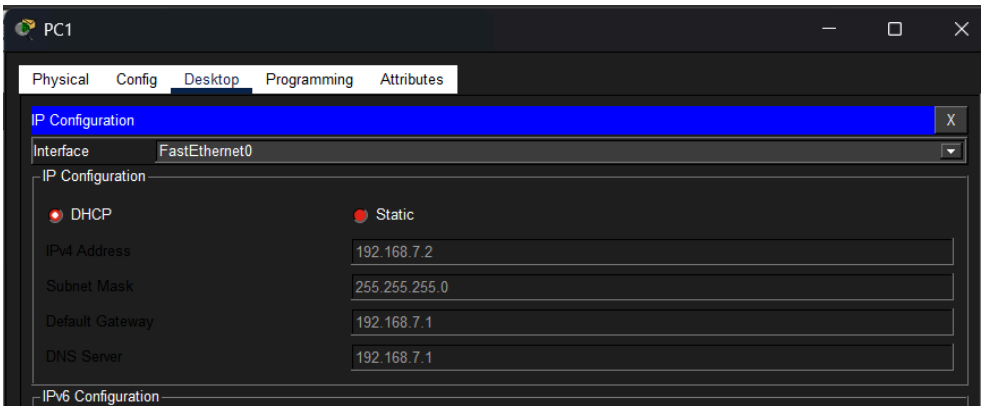
PC showing DHCP assigned IP:





PC3
Physical
Config
Desktop
Programming
Attributes
IP Configuration
X
Interface
FastEthernet0
IP Configuration
DHCP
Static
IPv4 Address
192.168.5.2
Subnet Mask
255.255.255.0
Default Gateway
192.168.5.1
DNS Server
192.168.5.1

PC0
Physical
Config
Desktop
Programming
Attributes
IP Configuration
X
Interface
FastEthernet0
IP Configuration
DHCP
Static
IPv4 Address
192.168.6.2
Subnet Mask
255.255.255.0
Default Gateway
192.168.6.1
DNS Server
192.168.6.1


PC1
Physical
Config
Desktop
Programming
Attributes
IP Configuration
X
Interface
FastEthernet0
IP Configuration
DHCP
Static
IPv4 Address
192.168.7.2
Subnet Mask
255.255.255.0
Default Gateway
192.168.7.1
DNS Server
192.168.7.1
IPv6 Configuration

PC2
Physical
Config
Desktop
Programming
Attributes
IP Configuration
X
Interface
FastEthernet0
IP Configuration
DHCP
Static
IPv4 Address
192.168.8.2
Subnet Mask
255.255.255.0
Default Gateway
192.168.8.1
DNS Server
192.168.8.1

OSPF configuration:

F1-Router:

```
F1-router#show running-config | section ospf
router ospf 10
 log-adjacency-changes
 network 10.10.10.8 0.0.0.3 area 0
 network 192.168.8.0 0.0.0.255 area 0
 network 192.168.7.0 0.0.0.255 area 0
 network 192.168.6.0 0.0.0.255 area 0
 network 10.10.105.4 0.0.0.3 area 0
```

F2-Router:

```
F2-router#show running-config | section ospf
router ospf 10
 log-adjacency-changes
 network 10.10.10.0 0.0.0.3 area 0
 network 192.168.3.0 0.0.0.255 area 0
 network 192.168.4.0 0.0.0.255 area 0
 network 192.168.5.0 0.0.0.255 area 0
 network 10.10.10.8 0.0.0.3 area 0
```

F3-Router:

```
F3-Router#show running-config | section ospf
router ospf 10
 log-adjacency-changes
 network 192.168.1.0 0.0.0.255 area 0
 network 192.168.2.0 0.0.0.255 area 0
 network 10.10.10.0 0.0.0.3 area 0
 network 10.10.10.4 0.0.0.3 area 0
F3-Router#
```

Routing table (show ip route):

F1-Router:

```
F1-router#show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

Gateway of last resort is not set

     10.0.0.0/8 is variably subnetted, 5 subnets, 2 masks
C       10.10.10.0/30 is directly connected, Serial0/3/0
L       10.10.10.2/32 is directly connected, Serial0/3/0
C       10.10.10.4/30 is directly connected, Serial0/3/1
L       10.10.10.6/32 is directly connected, Serial0/3/1
O       10.10.10.8/30 [110/192] via 10.10.10.1, 00:13:00, Serial0/3/0
O    192.168.1.0/24 [110/65] via 10.10.10.1, 00:55:01, Serial0/3/0
O    192.168.2.0/24 [110/65] via 10.10.10.1, 00:55:01, Serial0/3/0
O    192.168.3.0/24 [110/129] via 10.10.10.1, 00:55:01, Serial0/3/0
O    192.168.4.0/24 [110/129] via 10.10.10.1, 00:55:01, Serial0/3/0
O    192.168.5.0/24 [110/129] via 10.10.10.1, 00:55:01, Serial0/3/0
     192.168.6.0/24 is variably subnetted, 2 subnets, 2 masks
```

F2-Router:

```
F2-router#show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

Gateway of last resort is not set

     10.0.0.0/8 is variably subnetted, 5 subnets, 2 masks
O       10.10.10.0/30 [110/128] via 10.10.10.5, 4294967273:4294967291:4294967295, Serial0/3/0
C       10.10.10.4/30 is directly connected, Serial0/3/0
L       10.10.10.6/32 is directly connected, Serial0/3/0
C       10.10.10.8/30 is directly connected, Serial0/3/1
L       10.10.10.10/32 is directly connected, Serial0/3/1
O    192.168.1.0/24 [110/65] via 10.10.10.5, 4294967273:4294967291:4294967295, Serial0/3/0
O    192.168.2.0/24 [110/65] via 10.10.10.5, 4294967273:4294967291:4294967295, Serial0/3/0
     192.168.3.0/24 is variably subnetted, 2 subnets, 2 masks
C       192.168.3.0/24 is directly connected, GigabitEthernet0/0.30
L       192.168.3.1/32 is directly connected, GigabitEthernet0/0.30
     192.168.4.0/24 is variably subnetted, 2 subnets, 2 masks
 --More--
```

F3-Router:

```
F3-Router#show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

Gateway of last resort is not set

     10.0.0.0/8 is variably subnetted, 5 subnets, 2 masks
C       10.10.10.0/30 is directly connected, Serial0/3/1
L       10.10.10.1/32 is directly connected, Serial0/3/1
C       10.10.10.4/30 is directly connected, Serial0/3/0
L       10.10.10.5/32 is directly connected, Serial0/3/0
O       10.10.10.8/30 [110/128] via 10.10.10.6, 00:13:06, Serial0/3/0
     192.168.1.0/24 is variably subnetted, 2 subnets, 2 masks
C       192.168.1.0/24 is directly connected, GigabitEthernet0/0.10
L       192.168.1.1/32 is directly connected, GigabitEthernet0/0.10
     192.168.2.0/24 is variably subnetted, 2 subnets, 2 masks
C       192.168.2.0/24 is directly connected, GigabitEthernet0/0.20
L       192.168.2.1/32 is directly connected, GigabitEthernet0/0.20
 --More--
```

Task 7: Wireless Network Configuration

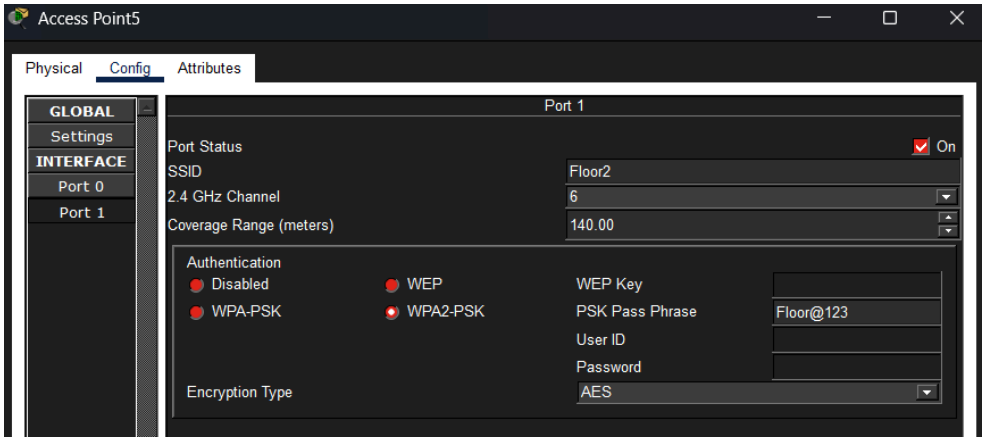
Problem:

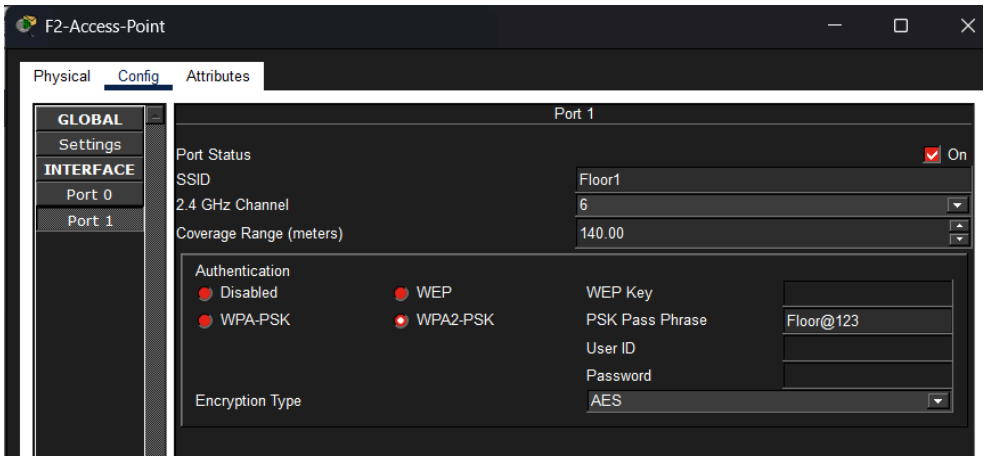
Provide Wi-Fi connectivity.

How we did:

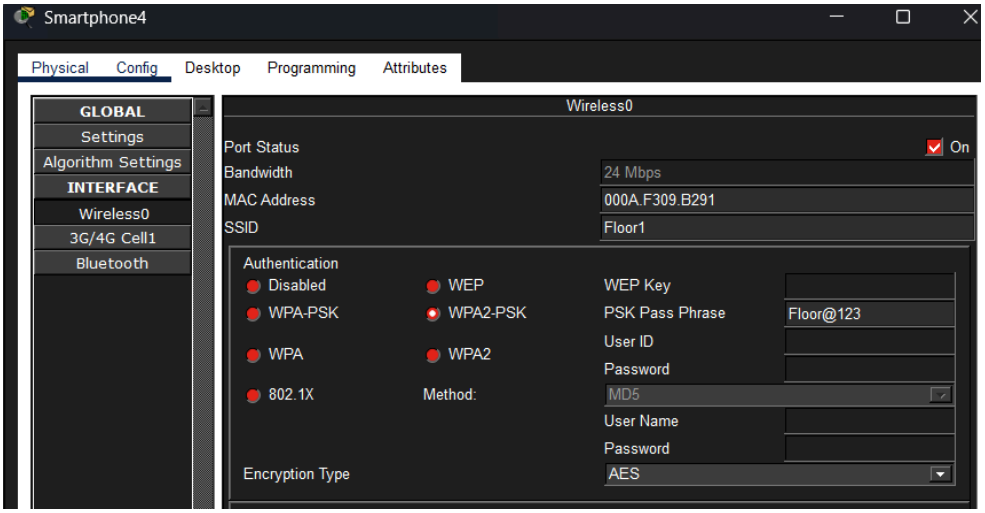
- Configured Access Points on each floor
- Set SSID and password
- Connected laptops, phones, and tablets

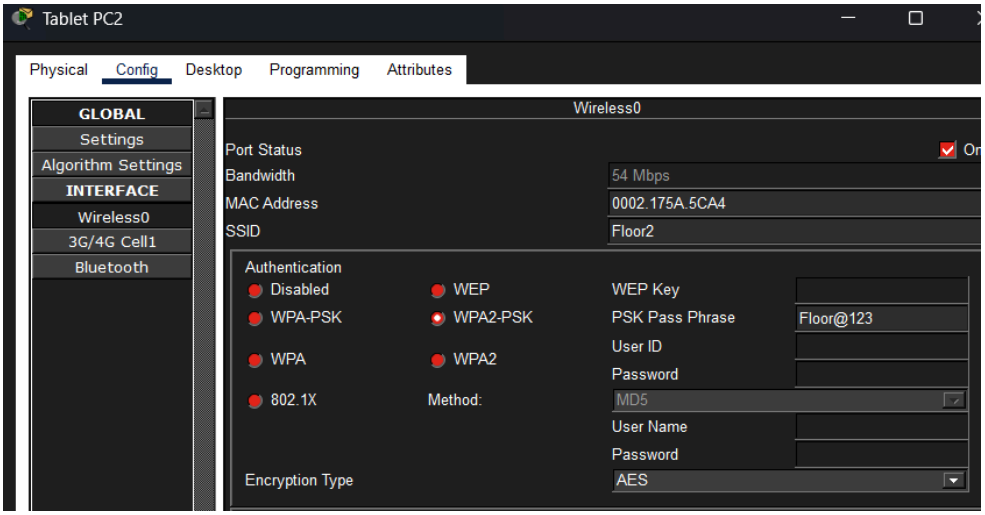
Access point configuration:





Wireless device connected:





Task 8: Configure SSH

Problem:

Enable secure remote login.

How we did:

hostname F1-Router

ip domain-name gtex

username admin password admin

crypto key generate rsa

1024

line vty 0 15

login local

transport input ssh

SSH configuration:

```
F3-Router#
F3-Router#show running-config | section vty
line vty 0 4
 login local
 transport input ssh
line vty 5 15
 login local
 transport input ssh
F3-Router#show running-config | include username
username getech password 0 getech
F3-Router#show ip ssh
SSH Enabled - version 1.99
Authentication timeout: 120 secs; Authentication retries: 3
F3-Router#
```

SSH login from PC

```
C:\>ssh -l gtech 192.168.6.1

Password:
% Login invalid


Password:



F1-router>

[Connection to 192.168.6.1 closed by foreign host]
C:\>
```

Task 9: Port Security Configuration

Problem:

Allow only one PC (Test PC) on IT switch port.

How we did:

interface fa0/2

switchport port-security

switchport port-security maximum 1

switchport port-security mac-address sticky

switchport port-security violation shutdown

Port security configuration:

```
!
!
!
!
!
spanning-tree mode pvst
spanning-tree extend system-id
!
interface FastEthernet0/1
 switchport mode trunk
!
interface FastEthernet0/2
 switchport access vlan 10
 switchport mode access
 switchport port-security
 switchport port-security mac-address sticky
 switchport port-security mac-address sticky 0060.5C1A.03B4
!
```

Show port-security output:

```
Switch#show port-security interface fa0/2
Port Security              : Enabled
Port Status                : Secure-up
Violation Mode             : Shutdown
Aging Time                 : 0 mins
Aging Type                 : Absolute
SecureStatic Address Aging : Disabled
Maximum MAC Addresses      : 1
Total MAC Addresses        : 1
Configured MAC Addresses   : 0
Sticky MAC Addresses       : 1
Last Source Address:Vlan   : 0060.5C1A.03B4:10
Security Violation Count   : 0

Switch#
```

Task 10: Verify Connectivity

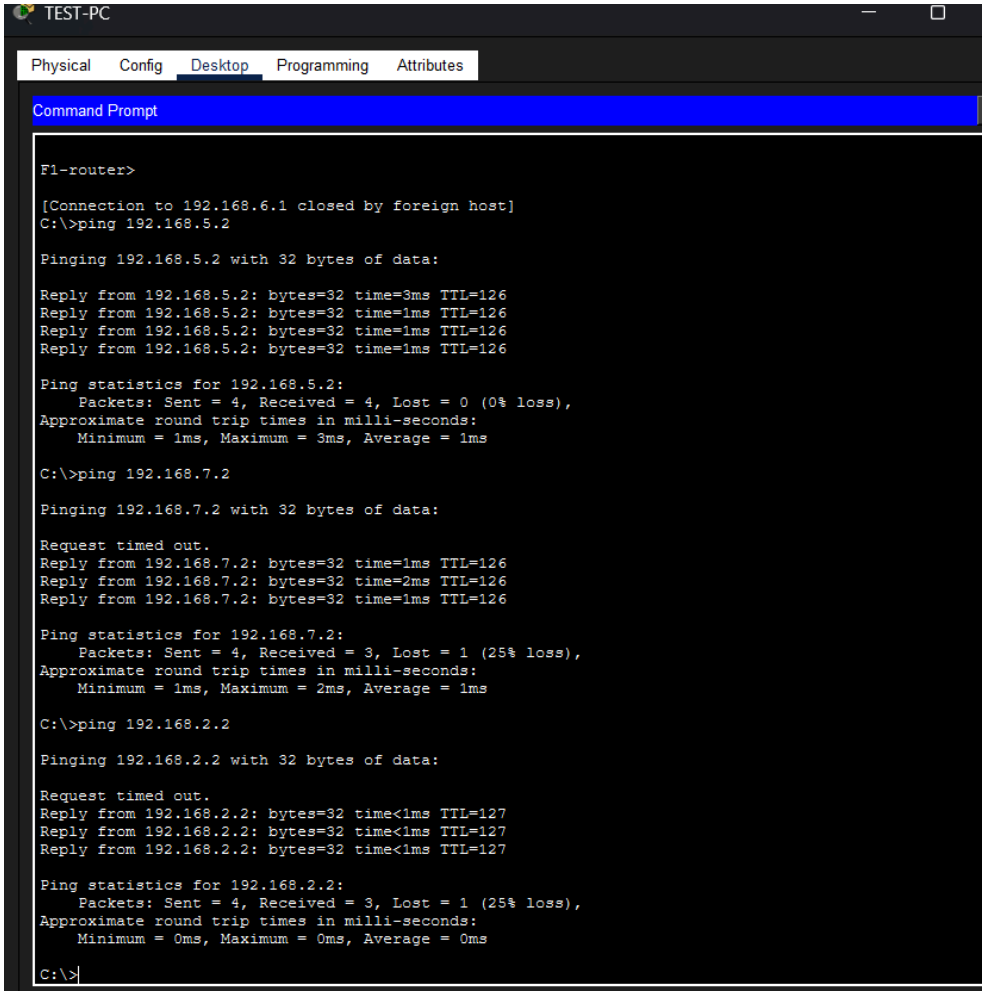
Problem:

Ensure all devices can communicate.

How we did:

- Used ping command
- Verified inter-VLAN communication
- Verified inter-floor communication using OSPF
- Tested SSH login from Test PC

Successful ping across floors:



REFLECTIONS:

1. Why are VLANs used?

VLANs are used to logically separate departments and improve network security and efficiency.

2. Why is Inter-VLAN routing required?

Because devices in different VLANs cannot communicate without routing.

	3. What is the role of OSPF? OSPF enables routers to dynamically share routing information and find the best path. **4. Why is DHCP used?** DHCP automatically assigns IP addresses, reducing manual configuration. **5. Why is SSH preferred over Telnet?** SSH provides encrypted communication, making it more secure than Telnet.

CONCLUSION	The experiment was successfully performed using Cisco Packet Tracer. A complex multi-floor network was designed and implemented with VLAN segmentation, inter-VLAN routing, DHCP, and OSPF routing. All devices were able to communicate across VLANs and floors. Wireless connectivity was established, and secure remote access was implemented using SSH. Port security ensured controlled access to critical network ports. This experiment provided a strong understanding of advanced networking concepts and real-world network design.